
Models Know, But Don't Always Say: A Guided Reproduction of Two Results on Hidden Knowledge in Language Models

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Abstract

A language model's output need not reflect what it internally represents. We reproduce two published results that make this gap concrete. First, following Marks & Tegmark (COLM 2024), we show that LLaMA-2-13B represents the truth of simple factual statements as a linearly accessible direction: true and false statements separate into two clusters under PCA, and linear probes trained on one dataset of statements generalize to others. Second, following Cywiński et al. (2025), we study *Taboo* models—Gemma-2-9B models fine-tuned to use a secret word downstream while never uttering it—and read the secret out of intermediate layers with the logit lens. All results are reproductions of published work with public code and models, performed with inference-only compute. The running theme: *the model knows, but does not necessarily say*—and simple tools from an introductory AI course suffice to see it.

1 Introduction

When a language model asserts a falsehood fluently, does any part of it “know” better? And when a model is explicitly trained to hide a piece of knowledge, does the knowledge stay hidden on the inside? Both questions ask about the gap between what a model *says* and what it *represents*—a gap with direct consequences for AI safety, where monitors that read only a model's text output can miss what the model is actually computing.

This report reproduces two published answers, chosen because they are striking, complementary, and achievable with the tools of an introductory AI course (PCA, logistic regression, linear discriminant analysis):

- **Reproduction A (main).** Marks & Tegmark [Marks and Tegmark, 2024] showed that LLaMA-2 models represent the truth of simple factual statements along linear directions: true and false statements cluster separately under PCA, probes generalize across datasets, and adding the “truth direction” to activations causally flips the model's answers.
- **Reproduction B (secondary).** Cywiński et al. [Cywiński et al., 2025] released *Taboo* models that know a secret word, use it, but never say it. White-box methods (logit lens on intermediate layers) recover the secret far more often than black-box prompting.

We re-run both pipelines end-to-end—activation extraction, probing, and interventions for A; logit-lens extraction for B—on the authors' public code and models, and report where our numbers match the originals. Everything here is a reproduction: each result is labeled with the figure or table of the original paper it corresponds to.

2 Background: reading activations

A decoder-only transformer [Touvron et al., 2023, Gemma Team, 2024] maintains, after each block, a *residual stream*: one d -dimensional vector per token position ($d = 5120$ for LLaMA-2-13B). If a property of the input (e.g. truthfulness) is linearly decodable from these vectors, we say the model *linearly represents* it [Alain and Bengio, 2017].

Probes. We use the three probe families of Marks and Tegmark [2024], all bias-free on mean-centered activations: (i) **LR**, logistic regression trained with AdamW; (ii) **MM** (mass-mean), the closed-form mean-difference direction $\theta = \mu^+ - \mu^-$, whose covariance-whitened variant is Fisher’s linear discriminant; (iii) **CCS** [Burns et al., 2023], an unsupervised contrastive probe.

Logit lens. The logit lens [nostalgebraist, 2020] applies the model’s final layer norm and unembedding to intermediate residual streams, reading off which vocabulary tokens the intermediate state already “prefers”. We use it to ask, layer by layer: *where in the network does the hidden knowledge become visible?*

3 Reproduction A: the geometry of truth

3.1 Setup

We follow Marks and Tegmark [2024] (arXiv:2310.06824v3; COLM 2024) and their public repository. **Model:** LLaMA-2-13B [Touvron et al., 2023], bfloat16, inference only. **Datasets:** the repository’s true/false statement datasets—e.g. `cities` (“The city of c is in C .”, 1,496 statements), `neg_cities` (negated variant), `sp_en_trans` (Spanish–English translations), and eight others with fixed templates and known ground truth. **Protocol:** raw declarative statements (no QA formatting, no few-shot priming); we extract the residual stream at the *final token* (the sentence-ending period) at 12 layers spanning the model, with per-dataset mean centering; probes are trained on a random 80/20 split (seed 0). Our activation extractor is a plain-transformers reimplementation that writes the exact file layout of the original repository (the upstream script depends on an outdated `nnsight` API); the probe training code is the authors’ own.

3.2 Result 1: true and false statements separate under PCA

Figure pending: PCA of last-token activations (LLaMA-2-13B, layer 14), colored by ground-truth label. Activations are being extracted on the cluster; this panel reproduces the style of Fig. 1 of the original paper.

Figure 1: **Two clusters without any labels.** Each point is one statement’s last-token activation projected onto the top two principal components (PCA is unsupervised). True and false statements separate geometrically. Reproduces the Fig. 1 analysis of Marks and Tegmark [2024].

Figure 1 shows the headline phenomenon: without ever seeing labels, PCA on mid-layer activations separates true from false statements into two clusters. Truth is therefore *linearly accessible* in the representation.

3.3 Result 2: probes generalize across datasets (*in progress*)

We train LR, MM, and CCS probes on one (combination of) dataset(s) and evaluate on all eleven, reproducing the generalization matrix of Fig. 5 of the original paper. Expected landmarks from the original: probes trained on `larger_than+smaller_than` exceed 95% accuracy on `sp_en_trans`; probes trained on `cities` alone fail on `neg_cities` (the two truth directions are nearly orthogonal); adding negated data to training fixes the failure.

3.4 Result 3: pushing along the truth direction flips answers (*in progress*)

Following §6 and Table 2 of the original: normalize the probe direction θ so that adding it moves the mean false statement onto the mean true one from the probe’s point of view, add it to the residual stream at layers 8–14 at the final tokens, and measure the normalized indirect effect (NIE) on $P(\text{TRUE}) - P(\text{FALSE})$ under a few-shot TRUE/FALSE prompt. The original finds MM directions strongly causal on 13B (NIE up to 0.90) while LR directions are far weaker (0.13–0.19) despite equal classification accuracy—*classification accuracy does not imply causal use*.

4 Reproduction B: reading a secret the model never says

(*In progress; scheduled after the midterm presentation.*) Cywiński et al. [2025] trained *Taboo* models: LoRA adapters on `gemma-2-9b-it` [Gemma Team, 2024] that play a taboo game—they know a secret word (e.g. *gold, moon, cloud*), use it to answer hint questions, and never utter it. We apply the logit lens to the residual stream at every layer while the model answers hint prompts, and record the rank of the secret word in each layer’s next-token distribution. The original paper reports that white-box access lifts an auditor’s top-1 accuracy on Taboo from 42% (text only) to 84% (logit-lens tokens), while the base model without the secret shows no such signal (19%); black-box attacks on Taboo stay below 8%—the secret is hidden from the output, but not from the activations.

5 Discussion and limitations

Everything in this report is a reproduction of published work; our contribution is pedagogical—showing that frontier interpretability results are reachable with course-level tools and inference-only compute. The statements are simple and template-based; truth geometry is richer than a single direction (negated and comparative datasets have near-orthogonal, even antipodal, directions at this scale); and all methods here require white-box access to weights, so they do not transfer to closed API models.

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This report reproduces published results: Reproduction A follows Marks and Tegmark [2024] (code: <https://github.com/saprm/geometry-of-truth>); Reproduction B follows Cywiński et al. [2025] (code and models: <https://github.com/cywinski/eliciting-secret-knowledge>). All original credit belongs to those authors; any errors in the reproduction are ours.

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